Lecture 14

The Normal and Exponential Distributions

The Normal (Gaussian) Distribution

- $X \sim N(\mu, \sigma^2)$ if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty.$$

- $E(X) = \mu$; $\text{Var}(X) = \sigma^2$.

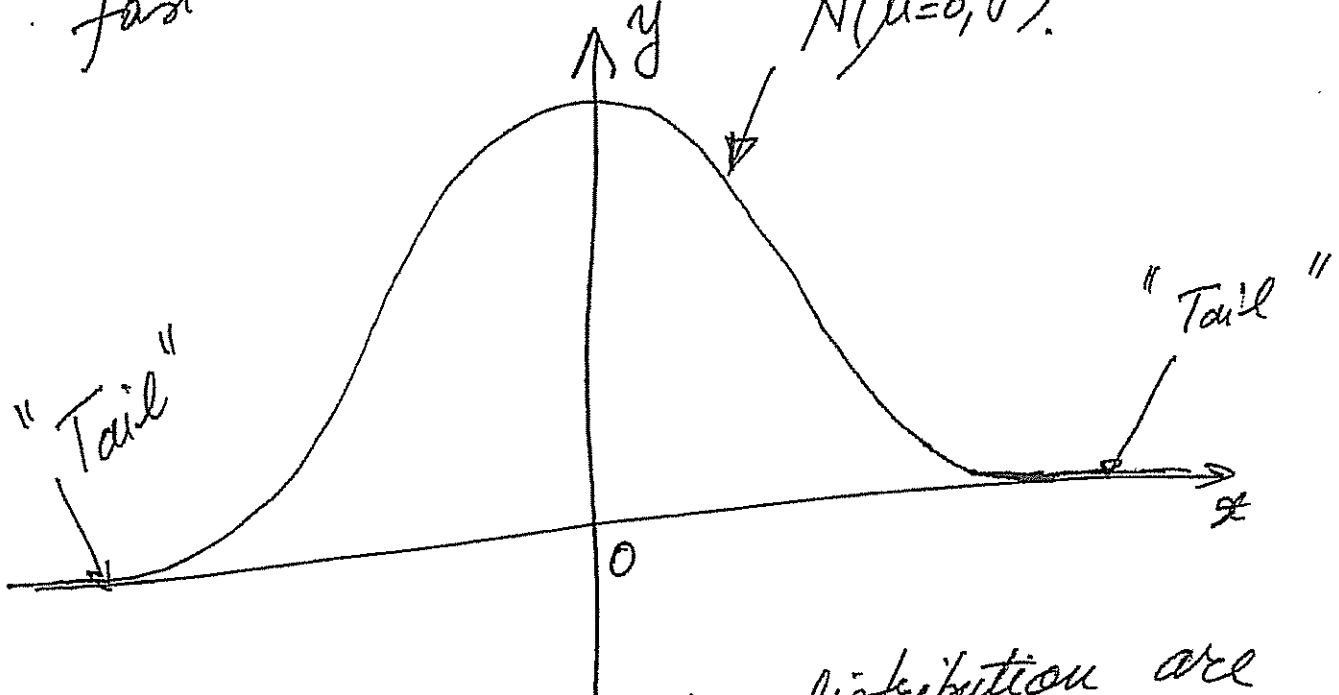
- μ = "Location parameter"
  σ = "Scale parameter"

- $\text{Median}(X) = \mu$.
- $f_X(x)$ is symmetric with respect to μ.

Result If $X \sim N(\mu, \sigma^2)$, then

$$Y =: aX+b \sim N(a\mu+b,\ a^2\sigma^2).$$

NB The difference between the Gaussian distribution and any other "bell shaped" distribution is in how fast its "tails" approach the Ox axis.



- The "tails" of a Gaussian distribution are "very light". In contrast, those of a Cauchy distr. (which does not even have a mean) are very "heavy".

The standard normal distribution

- $X \sim N(\mu = 0, \sigma^2 = 1)$;

(1) $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

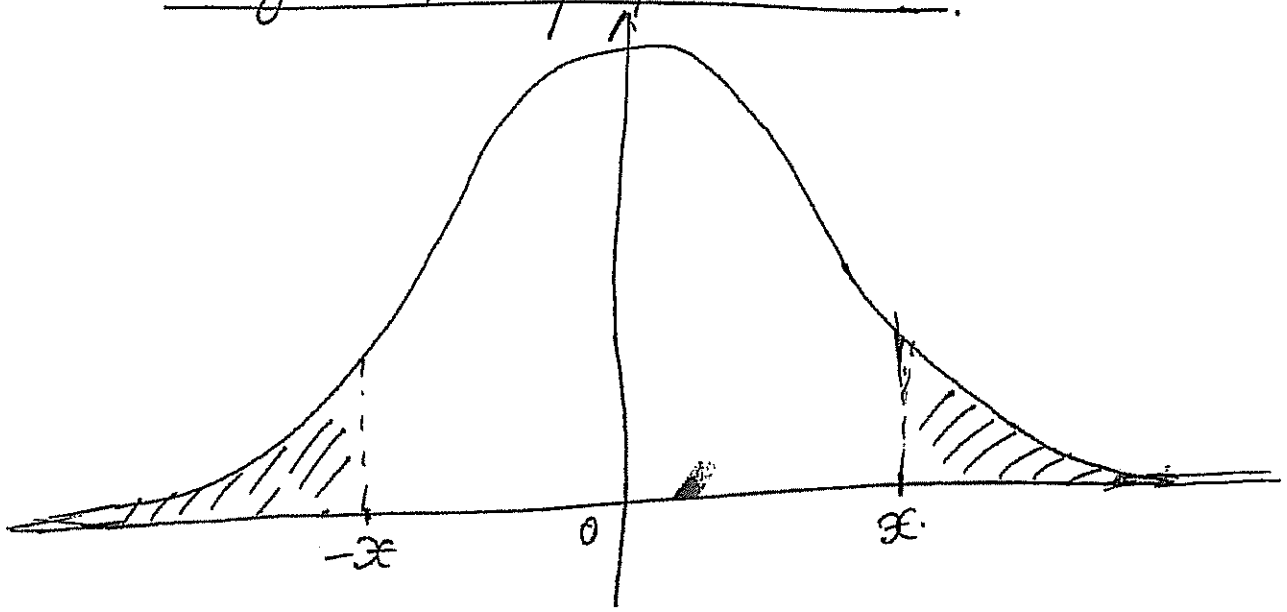
(2) $\Phi_X(x) =: \Phi(x) =: \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$.

The cumulative distribution function of a standard normal has a dedicated letter, Φ.

- No closed form for $\Phi(x)$; use tables!

- Recall: $\Phi(x) = \mathbb{P}(X \leq x)$.

Very useful properties



(1) $\Phi(-x) = 1 - \Phi(x)$

(2) If $X \sim N(\mu, \sigma^2)$ then

$$Z =: \frac{X-\mu}{\sigma} \sim N(0,1).$$

(3) If $X \sim N(\mu, \sigma^2)$ then

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

(4) $\mathbb{P}(-2 \le Z \le 2) = 2\,\Phi(2) - 1 = 2 \times 0.97 - 1 = 0.94$

$\mathbb{P}(-3 \le Z \le 3) = 2\,\Phi(3) - 1 = 2 \times 0.998 - 1 = 0.996$

General rule for computing $P(a < X < b)$, for $X \sim N(\mu, \sigma^2)$, using the Table at page 861

$$\mathbb{P}(a < X < b) = \mathbb{P}\left(\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right)$$

$$= \mathbb{P}\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right)$$

$\frac{X-\mu}{\sigma} \overset{\text{definition}}{=\!=} Z$

$$= \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$$

our notation

Example

If $X \sim N(\mu = 10, \sigma^2 = 36)$ find

$\mathbb{P}(4 < X < 16)$. //

Solution $\mathbb{P}(4 < X < 16) =$

$$= \Phi\left(\frac{16-10}{6}\right) - \Phi\left(\frac{4-10}{6}\right)$$

$$= \Phi(1) - \Phi(-1)$$

$$= \Phi(1) - [1 - \Phi(+1)]$$

$$= 2 \cdot \Phi(1) - 1$$

$$= 2 \cdot 0.84 - 1 = 0.68$$ //

$=$ ↑ Table

• See Fig. 5.4 and Table 5.2 for comparisons between normal distributions with different parameters.

Linear combinations of Gaussian variables

Result 1 If $X_i \sim N(\mu_i, \sigma_i^2)$, $1 \le i \le k$, are k *independent* Gaussian variables and if $a_1, \ldots, a_k, b$ are given constants, then $Z = a_1 X_1 + a_2 X_2 + \ldots + a_k X_k + b$ has the following normal distribution:

$$N\left(\mu = \sum_{i=1}^{k} a_i \mu_i + b, \sigma^2 = \sum_{i=1}^{k} a_i^2 \sigma_i^2\right).$$

Example (The Sample mean).

- Let $X_1, \ldots, X_n$ be n independent $N(\mu, \sigma^2)$ random variables. (This means that they <u>all</u> have this distribution).
- Define their "sample mean" $\overline{X}_n$ as:

$$\overline{X}_n = \frac{X_1 + X_2 + \ldots + X_n}{n} = \frac{\sum_{i=1}^{n} X_i}{n}.$$

- Then, $\overline{X}_n = \frac{1}{n} \cdot X_1 + \frac{1}{n} X_2 + \ldots + \frac{1}{n} X_n$ and, by Result 1, we have that:

$$\overline{X}_n \sim N\left(\mu_n = \sum_{i=1}^{n} \frac{\mu}{n},\ \sigma_n^2 = \sum_{i=1}^{n} \frac{\sigma^2}{n}\right),$$

that is $\boxed{\overline{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$ (1).

Important Application (Determining the sample size)

(How many measurements to take in order to be sure that $\bar{X}_n$ is "close" to μ with some (given) high probability ?)

- Let $X \sim N(\mu, \sigma^2 = 9)$.
  Let $X_1, \ldots, X_n$ be n independent measurements on X. Find the smallest value of n such that

$$\mathbb{P}(|\bar{X}_n - \mu| \leq 1) \geq 0.95$$

//

Remark You will see that we don't need to know the value of μ to solve this problem.

Solution

- We know that

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right), \text{ for us } \sigma^2 = 9, \text{ so}$$

$$\bar{X}_n \sim N\left(\mu, \frac{9}{n}\right)$$

Therefore

$$Z =: \frac{\bar{X}_n - \mu}{\sqrt{\frac{9}{n}}} \sim N(0,1),$$

and so

$$\mathbb{P}\left(|\bar{X}_n - \mu| \leq 1\right) = \mathbb{P}\left(\left|\frac{\bar{X}_n - \mu}{\sqrt{\frac{9}{n}}}\right| \leq \frac{1}{\frac{3}{\sqrt{n}}}\right)$$

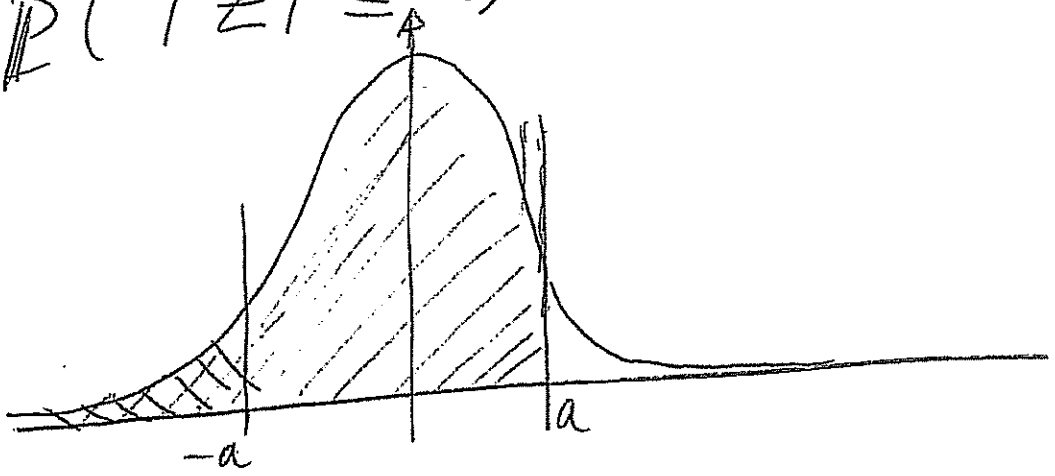
$$= \mathbb{P}\left(|Z| \leq \frac{\sqrt{n}}{3}\right).$$

Thus, we need to find the smallest n such that:

$$\mathbb{P}\left(|Z| \leq \frac{\sqrt{n}}{3}\right) \geq 0.95.$$

- Consider a, a fixed value. Then

$$\textcircled{*} = \mathbb{P}(|Z| \leq a) = \mathbb{P}(-a \leq Z \leq a).$$



$$\textcircled{*} = \Phi(a) - \Phi(-a) = \Phi(a) - (1 - \Phi(a))$$
$$= 2\Phi(a) - 1.$$

• Therefore, for $a = \frac{\sqrt{n}}{3}$ we have:

$$\mathbb{P}\left(|Z| \leq \frac{\sqrt{n}}{3}\right) = 2\Phi\left(\frac{\sqrt{n}}{3}\right) - 1,$$

So we need to find n such that

$$2\Phi\left(\frac{\sqrt{n}}{3}\right) - 1 \geq 0.95$$

$$\Updownarrow$$

$$\Phi\left(\frac{\sqrt{n}}{3}\right) \geq \frac{1.95}{2}$$

$$\Updownarrow$$

$$\Phi\left(\frac{\sqrt{n}}{3}\right) \geq 0.975$$

$$\Updownarrow \longleftarrow \text{From Table page 861.}$$

$$\frac{\sqrt{n}}{3} \geq 1.96 \iff \boxed{n \geq 35}.$$

q.e.d.

The Exponential Distribution

Def $X \sim \text{Exponential}(\beta)$ if

$$f_X(x) = \begin{cases} \beta e^{-\beta x}, & \text{for } x > 0, \\ 0, & \text{for } x \leq 0. \end{cases}$$

- Typically used to model lifetimes of devices.

$$E(X) = \beta \int_0^\infty x \cdot e^{-\beta x} dx, \quad (1)$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2, \text{ where}$$

$$E(X^2) = \beta \int_0^\infty x^2 \cdot e^{-\beta x} dx. \quad (2)$$

Remark The Exponential Distribution is a special case of a general class of distributions called the Gamma Distributions (page 319), obtained for $\alpha = 1$.

- To help calculate integrals like (1) and (2) we introduce a general function that has them as special cases.

The Gamma function

For any $\alpha > 0$, define

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx, \text{ called}$$

the gamma function.

Results (1) For any $\alpha > 1$: $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$.

(2) For any integer n: $\Gamma(n) = (n-1)!$

(3) For each $\alpha > 0$ and $\beta > 0$:

$$\boxed{\int x^{\alpha-1} \cdot e^{-\beta x}\, dx = \frac{\Gamma(\alpha)}{\beta^\alpha}}$$

- By Result (3):

$$E(X) = \beta \cdot \frac{\Gamma(2)}{\beta^2} = \frac{1}{\beta} \cdot \Gamma(2) \Longrightarrow$$

By Result 2: $\Gamma(2) = 1!$

$$\boxed{E(X) = \frac{1}{\beta}, \text{ for } X \sim \text{Exponential}(\beta)}$$

- $E(X^2) = \beta \cdot \frac{\Gamma(3)}{\beta^3} = \frac{2!}{\beta^2} = \frac{2}{\beta^2}$.

Therefore: $\boxed{\text{Var}(X) = \frac{2}{\beta^2} - \frac{1}{\beta^2} = \frac{1}{\beta^2}, \text{ for } X \text{ as above.}}$

The exponential random variable

(Summary)

- $X \sim f(x)$

$$f(x) = \begin{cases} \beta e^{-\beta x} & ; \text{ for } x \geq 0 \\ 0 & ; \text{ for } x < 0. \end{cases}$$, for some $\beta > 0$.

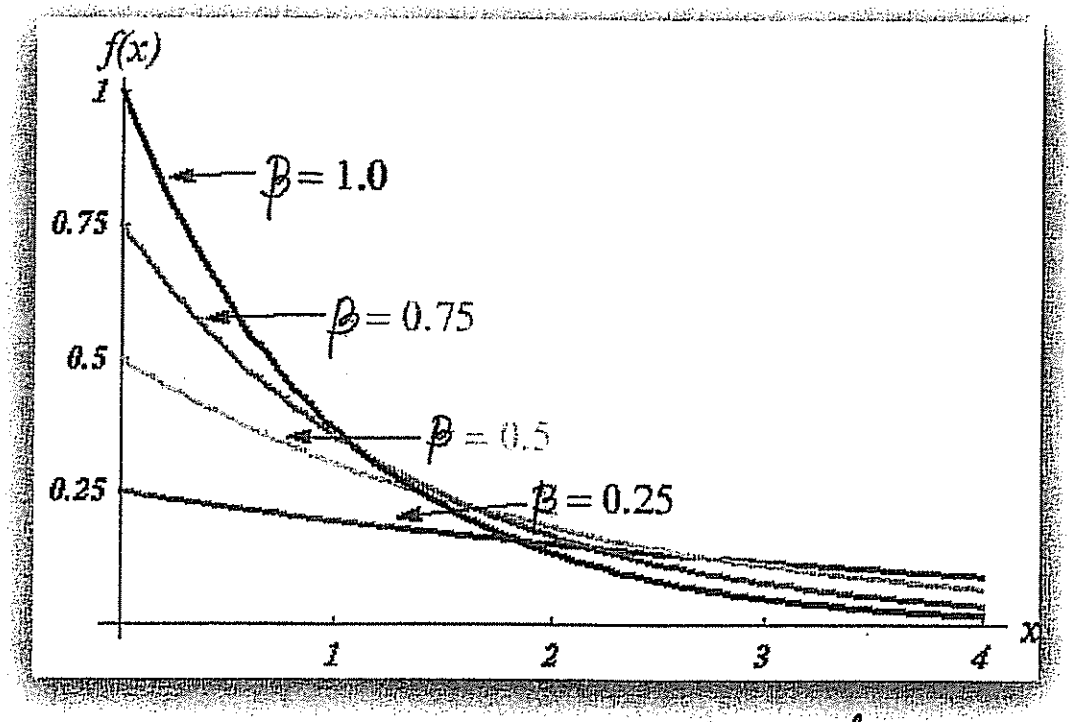
- $F(x) = P(X \leq x) = 1 - e^{-\beta x}$; $x \geq 0$.
- $EX = \frac{1}{\beta}$
- $\text{var } X = \frac{1}{\beta^2}$

Definition A r.v. X is called memoryless if

- $P(X > s+t \mid X > t) = P(X > s)$, for all $s, t \geq 0$.

Proposition 1 X is memoryless if and only if X has an exponential distribution.

Plot of the p.d.f. of an Exponential (β) distribution for various values of β.



NOTE : The larger the value of β, the steeper the descent of the curve.

Example 1

X = # of years a radio functions.

$X \sim$ Exponential $(\beta = \frac{1}{8})$.

Say the radio has been used for t years.

Then, we need

$P(X > t+8 \mid X > t)$. Because X is memoryless, by Proposition 1 we have

$$P(X > t+8 \mid X > t) = P(X > 8) =$$

$$= \frac{1}{8} \int_8^\infty e^{-\frac{x}{8}} dx = \frac{1}{8} \cdot (-8)\, e^{-\frac{x}{8}} \Big|_8^\infty =$$

$$= -e^{-\frac{\infty}{8}} + e^{-\frac{8}{8}} = 0 + e^{-1} = e^{-1}.$$

Example 2. Used your car for 10K miles. Can use for 20K more.

X = # of thousands of miles before throwing the car away.

a) $X \sim$ Exponential $\left(\frac{1}{20}\right)$.

$\therefore P(X > 20 + 10 \mid X > 10) \overset{\text{memoryless}}{=} P(X > 20)$

$$= \frac{1}{20}\int_{20}^{\infty} e^{-\frac{x}{20}}\,dx = e^{-1} \simeq 0.3678$$

b) Assume now that $X \sim$ Unif $([0, 40])$.

Then $P(X > 30 \mid X > 10) = \dfrac{P(X > 30)}{P(X > 10)}$

$$P(X > 30) = \frac{1}{40}\int_{30}^{40} dx = \frac{10}{40} = \frac{1}{4}$$

$$P(X > 10) = \frac{1}{40}\int_{10}^{40} dx = \frac{30}{40} = \frac{3}{4}$$

So $P(X > 30 \mid X > 10) = \dfrac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3} \simeq 0.33$.